

BATHROOM LIGHT

In the DIY article 'A simple Automatic Bathroom Light' published in May issue, on page no. 82, in first paragraph line no.10 begins with "a reed switch with normally open contacts as the sensor," but in fifth paragraph, line 4 begins with "a normally closed-type reed switch." Which is correct, normally open or normally closed relay?

Roy Vergheze

EFY. We thank Roy for detecting this error and sharing with everyone! Please read normally open as normally closed as normally closed-type reed switch has been used in the circuit.

ACRONYMS

In May issue, under Industry News item 'Bharat Electronics signs...' on page 60 (3rd line), IDMM is mentioned as the acronym for Indigenously Designed, Developed and Manufactured. The correct acronym is IDDM and not IDMM.

In same issue, in second paragraph on page 111, NPI is mentioned as an acronym for New Product Development. The acronym should be NPD instead of NPI.

Sachin Goel

EFY. We thank Sachin for detecting the wrong acronyms and sharing with everyone!

PROTECTION CIRCUIT

This is regarding my DIY article ‘Over-voltage and Undervoltage Protection for Electrical Appliances’ published in April issue. Please note, diodes D1 through D4 were shown wrongly

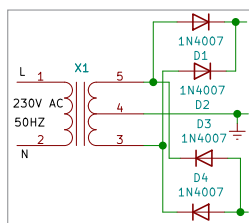


Fig. 1: Correct diode connections

connected in the circuit diagram. The correct connections are shown in Fig. 1 below. Sorry for any inconvenience to the readers!

Shamail

TRANSFORMER DESIGN

Please publish a detailed article on transformer core size vs its current output. I recently came across a reply from an author, which was recently published under this section in EFY, which said that transformers available in local markets do not give the rated current according to their core size. So, I will appreciate an article on this subject.

Anirvan Kule, Kolkata

EFY. Transformers are an essential part of many electrical systems. They are used to transmit power, distribute power, and convert voltages. By understanding the relationship between transformer core size and ampere output, you can select the right transformer for your application.

A transformer transfers energy from one electrical circuit to another without the use of moving parts. It is based on the principle of electromagnetic induction.

The transformer has two coils of wire, the primary coil and the secondary coil. The primary coil is connected to the input voltage source and the secondary coil is connected to the output load.

When an alternating current (AC) is applied to the primary coil, it creates a magnetic field. This magnetic field passes through the secondary coil and induces an AC voltage in the secondary coil. The voltage induced in the secondary coil is proportional to the number of turns in the secondary coil divided by the number of turns in the primary coil.

The transformer core is made of a ferromagnetic material, such as iron or steel. The core's purpose is to concentrate the magnetic field so that it is more effective in inducing a voltage

in the secondary coil. The core size is determined by the amount of current that the transformer will be carrying. The larger the core, the more current it can carry without saturating.

The current output of a transformer is determined by the cross-sectional area of the core, the frequency of the input voltage, and the number of turns in the secondary coil. The cross-sectional area of the core determines the amount of magnetic flux that can be produced.

The frequency of the input voltage determines the number of times per second that the magnetic field changes direction. The number of turns in the secondary coil determines the amount of voltage that is induced in the secondary coil.

The following formula can be used to calculate the ampere output of a transformer: $\text{Amperage} = (\text{Cross-sectional area of core}) \times (\text{Frequency of input voltage}) \times (\text{Number of turns in secondary coil})$

The transformer core size and the ampere output are directly related. The larger the core, the more current the transformer can carry. The smaller the core, the less current the transformer can carry.

It is important to select a transformer with a core size that is large enough to handle the current that the transformer will be carrying. If the core is too small, the transformer will overheat and could be damaged.

Here are some additional factors to consider when selecting a transformer:

Voltage. The transformer must be able to handle the input voltage and the output voltage.

Frequency. The transformer must be designed for the frequency of the input voltage.

Insulation. The transformer must be insulated to prevent electrical shock.

Cooling. The transformer must be able to dissipate heat without overheating.